Chapter 8. Functions Defined by Integrals

Continuity and differentiation under the integral sign, the gamma and beta functions, integral transforms

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8.1 Integrals depending on a parameter

Motivation. When is $x \mapsto \int f(x,t)\,dt$ continuous, and differentiable under the integral sign?

For f on $[a,b]\times[c,d]$ let $\phi(x)=\int_c^d f(x,t)\,dt$. Write $D_1 f=\partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f:[a,b]\times[c,d]\to\mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. (a) By uniform continuity (Theorem 3.18 of Volume I), $|f(x,t)-f(x',t)|<\epsilon$ for $|x-x'|<\delta$, all t, so $|\phi(x)-\phi(x')|\le\epsilon(d-c)$.

(b) $D_1 f$ is uniformly continuous on $[x-\eta,x+\eta]\times[c,d]\subset(a,b)\times[c,d]$. For $0<|h|<\eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h}=D_1 f(x+\theta h,t)$ with $0<\theta=\theta(h,t)<1$, so

$$\left|\frac{\phi(x+h)-\phi(x)}{h}-\int_c^d D_1 f(x,t)\,dt\right|\le\int_c^d |D_1 f(x+\theta h,t)-D_1 f(x,t)|\,dt\le\epsilon(d-c)$$

once $|h|$ is small. ■

Example 8.1

Differentiating $\int_0^1 t^x\,dt=\frac{1}{x+1}$ n times in $x>-1$ gives $\int_0^1 t^x(\log t)^n dt=(-1)^n n!/(x+1)^{n+1}$. Also $\int_a^b\phi(x)\,dx=\int_c^d\int_a^b f(x,t)\,dx\,dt$ for continuous f: as functions of b both vanish at $b=a$ with derivative $\int_c^d f(b,t)\,dt$ ((a), (b), Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d\le\infty)$, consider the improper integral $\phi(x)=\int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, and suppose there is $g:[c,d)\to\mathbb{R}$ with (a) $|f(x,t)|\le g(t)$ for all (x,t) and (b) $\int_c^d g<\infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence: comparison (Proposition 7.8). If $\int_{d'}^d g<\epsilon$, then $|\phi-\phi_{d'}|<\epsilon$ with $\phi_{d'}(x)=\int_c^{d'} f(x,t)\,dt$ continuous (Proposition 8.1); so ϕ is a uniform limit of continuous functions (Theorem 6.1). ■

Theorem 8.3 (8.1.3)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0\in(a,b)$. If there is $h:[c,d)\to\mathbb{R}$ with (a) $|D_1 f(x,t)|\le h(t)$ and (b) $\int_c^d h<\infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x\in(a,b)$, and ϕ is differentiable with $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. $|f(x,t)-f(x_0,t)|\le|x-x_0|h(t)$ (mean value theorem) gives convergence by comparison. $\phi_{d'}(x)=\int_c^{d'} f(x,t)dt$ has $\phi'_{d'}(x)=\int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), converging uniformly as $d'\to d$ (tails $\le\int_{d'}^d h$), and $\phi_{d'}(x_0)\to\phi(x_0)$; apply Theorem 6.4 along $d'_n\to d$. ■

Example 8.2

(a) $F(x)=\int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x)=-\int_0^\infty te^{-t^2}\sin(xt)\,dt=-\frac{x}{2}F(x)$ (integration by parts), so $F(x)=F(0)e^{-x^2/4}=\frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx=\frac{\pi}{2}$: for $G(s)=\int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s>0)$, Theorem 8.3 gives $G'(s)=-\frac{1}{1+s^2}$, and $G(s)\to 0$ as $s\to\infty$, so $G(s)=\frac{\pi}{2}-\arctan s$; continuity at $s=0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

Motivation. Γ extends the factorial; Section 8.1 gives its basic properties.

For $x>0$ let $\Gamma(x)=\int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t}\le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x)=\int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1)=x\Gamma(x)$, and $\Gamma(1)=1$, so $\Gamma(n+1)=n!$.

A function $f:I\to\mathbb{R}$ is **convex** if $f((1-\lambda)x+\lambda y)\le(1-\lambda)f(x)+\lambda f(y)$ for $x,y\in I$, $0\le\lambda\le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f:I\to\mathbb{R}$ with $f(\frac{x+y}{2})\le\frac{f(x)+f(y)}{2}$ for all $x,y\in I$ is convex.
(b) A convex f satisfies, for $s<t<u$ in I,

$$\frac{f(t)-f(s)}{t-s}\le\frac{f(u)-f(s)}{u-s}\le\frac{f(u)-f(t)}{u-t}.$$

Proof. (a) Induct on n for dyadic $\lambda=k/2^n$: if $\lambda=\frac{\lambda_1+\lambda_2}{2}$ with λ_i of level $n-1$, $(1-\lambda)x+\lambda y$ is the midpoint of the points for λ_1,λ_2. General λ: dyadic $\lambda_n\to\lambda$ and continuity.
(b) $t=(1-\lambda)s+\lambda u$, $\lambda=\frac{t-s}{u-s}$, so $f(t)\le(1-\lambda)f(s)+\lambda f(u)$; rearrange. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big)=\int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt\le\Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f:(0,\infty)\to(0,\infty)$ satisfies (a) $f(x+1)=xf(x)$, (b) $f(1)=1$, (c) $\log f$ is convex, then $f=\Gamma$.

Proof. Γ has (a)–(c), so we show they determine f, and by (a) on $(0,1]$ suffices. Let $\phi=\log f$, $0<x\le 1$; $\phi(n+1)=\log n!$. Proposition 8.4(b) at $n<n+1<n+1+x$ and $n+1<n+1+x<n+2$ gives

$$\log n\le\frac{\phi(n+1+x)-\phi(n+1)}{x}\le\log(n+1).$$

By (a), $\phi(n+1+x)=\phi(x)+\log[x(x+1)\cdots(x+n)]$. Hence

$$0\le\phi(x)-\log\frac{n!\,n^x}{x(x+1)\cdots(x+n)}\le x\log\Big(1+\frac{1}{n}\Big)\to 0,$$

so $f(x)=\lim_n\frac{n!\,n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ∎

Example 8.3

The **beta function** $B(x,y)=\int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y)=\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x)=\Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac12)^2=B(\frac12,\frac12)=\pi$, so $\int_{-\infty}^{\infty}e^{-x^2}dx=\Gamma(\frac12)=\sqrt{\pi}$. Stirling's formula $\Gamma(x+1)\sim\sqrt{2\pi x}\,x^xe^{-x}$ follows from the substitution $t=x(1+u)$ and Laplace's method.

8.3 Selected exercises

Motivation. Book exercises: classical integrals by differentiation under the integral sign, and Γ.

Exercise 8.4.1 (The Gaussian integral)

Let

$$f(x)=\Big(\int_0^x e^{-t^2}dt\Big)^2,\qquad g(x)=\int_0^1\frac{e^{-x^2(t^2+1)}}{t^2+1}\,dt.$$

Compute $(f+g)'$, show that $f(x)+g(x)=\frac{\pi}{4}$, and deduce $\lim_{A\to\infty}\int_0^A e^{-t^2}dt=\frac12\sqrt{\pi}$.

Solution.

Step 1. By the fundamental theorem of calculus, $f'(x)=2e^{-x^2}\int_0^x e^{-t^2}dt$.

Step 2. $h(x,t)=e^{-x^2(t^2+1)}/(t^2+1)$ and $D_1h=-2xe^{-x^2(t^2+1)}$ are continuous, so by Proposition 8.1(b)

$$g'(x)=-2e^{-x^2}\int_0^1 xe^{-x^2t^2}dt=-2e^{-x^2}\int_0^x e^{-u^2}du,$$

($u=xt$, any real x).

Step 3. So $(f+g)'=0$ and $f+g\equiv f(0)+g(0)=\int_0^1\frac{dt}{1+t^2}=\frac{\pi}{4}$.

Step 4. $0\le g(x)\le\frac{\pi}{4}e^{-x^2}\to0$, so $f(x)\to\frac{\pi}{4}$; taking square roots, $\int_0^A e^{-t^2}dt\to\frac{\sqrt{\pi}}{2}$. ∎

Exercise 8.4.4 (A Laplace-type integral)

For $x\in\mathbb{R}$ let

$$f(x)=\int_0^\infty\frac{\cos tx}{1+t^2}\,dt,\qquad g(x)=\int_0^\infty\frac{\sin tx}{t(1+t^2)}\,dt.$$

(a) Show that g is differentiable on $\mathbb{R}$ and $g'=f$.
(b) Show that f is continuous on $\mathbb{R}$.
(c) Show that f is differentiable on $(0,\infty)$ and $g''(x)-g(x)+\frac{\pi}{2}=0$ for $x>0$.
(d) Show that $f(x)=\frac{\pi}{2}e^{-|x|}$ for every $x\in\mathbb{R}$.

Solution.

(a) $k(x,t)=\frac{\sin tx}{t(1+t^2)}$ $(t>0)$, $k(x,0)=x$, is continuous ($\frac{\sin tx}{t}\to x$ locally uniformly), $|k|\le\frac{|x|}{1+t^2}$, and $D_1k=\frac{\cos tx}{1+t^2}$ is continuous with $|D_1k|\le\frac{1}{1+t^2}$ integrable. Theorem 8.3 gives $g'(x)=\int_0^\infty\frac{\cos tx}{1+t^2}dt=f(x)$.

(b) $\big|\frac{\cos tx}{1+t^2}\big|\le\frac{1}{1+t^2}$: f is continuous (Theorem 8.2), $|f|\le\frac{\pi}{2}=f(0)$.

(c) Fix $x>0$.

Step 1. $\frac{1}{t(1+t^2)}=\frac1t-\frac{t}{1+t^2}$ and $\int_0^\infty\frac{\sin tx}{t}\,dt=\frac{\pi}{2}$ (Example 8.2(b), $u=tx$), so $\int_0^\infty\frac{t\sin tx}{1+t^2}dt$ converges,

and $u = tx$ gives

$$g(x) = \frac{\pi}{2} - H(x), \qquad H(x) = \int_0^\infty \frac{u \sin u}{x^2 + u^2}\, du.$$

Step 2. Integrating by parts, with $\frac{d}{du}\frac{u}{x^2+u^2} = \frac{x^2-u^2}{(x^2+u^2)^2}$ and boundary terms $-\frac{u\cos u}{x^2+u^2}\big|_0^\infty = 0$,

$$H(x) = \int_0^\infty \frac{(x^2 - u^2)\cos u}{(x^2 + u^2)^2}\, du.$$

Step 3. Also $f(x) = \int_0^\infty \frac{x\cos u}{x^2+u^2}\, du$, and $\left|\frac{\partial}{\partial x}\frac{x\cos u}{x^2+u^2}\right| = \left|\frac{(u^2-x^2)\cos u}{(x^2+u^2)^2}\right| \le \frac{1}{a^2+u^2}$ for $x \ge a > 0$. Theorem 8.3 on each (a, b) and Step 2 give

$$f'(x) = \int_0^\infty \frac{(u^2 - x^2)\cos u}{(x^2 + u^2)^2}\, du = -H(x) = g(x) - \frac{\pi}{2}.$$

Step 4. With $g' = f$ from (a), $g''(x) = f'(x) = g(x) - \frac{\pi}{2}$.

(d)

Step 1. $y = g - \frac{\pi}{2}$ has $y'' = y$ on $(0, \infty)$; $w = y' + y$ has $w' = w$, so $w = Ce^x$, $(e^x y)' = Ce^{2x}$, and $y = Ae^x + Be^{-x}$.

Step 2. $f = y' = Ae^x - Be^{-x}$ is bounded (b), so $A = 0$; continuity at 0 and $f(0) = \frac{\pi}{2}$ give $-B = \frac{\pi}{2}$.

Step 3. So $f(x) = \frac{\pi}{2}e^{-x}$ for $x \ge 0$, and f is even. ∎

Exercise 8.4.6 (Gamma at half-integers)

Show that $\Gamma(n + \frac{1}{2}) = \frac{(2n)!}{n!\,4^n}\sqrt{\pi}$ for $n = 0, 1, 2, \ldots$.

Solution. For $n = 0$ this is $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (Example 8.3). If it holds for n, then $\Gamma(x+1) = x\Gamma(x)$ gives

$$\Gamma\Big(n + \frac{3}{2}\Big) = \frac{2n+1}{2} \cdot \frac{(2n)!}{n!\,4^n}\sqrt{\pi} = \frac{(2n+2)(2n+1)(2n)!}{4(n+1)\,n!\,4^n}\sqrt{\pi} = \frac{(2n+2)!}{(n+1)!\,4^{n+1}}\sqrt{\pi},$$

which is the claim for $n + 1$. ∎

Exercise 8.4.7 (A series for Γ)

Show that for every $x > 0$,

$$\Gamma(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \frac{1}{n + x} + \sum_{n=0}^{\infty} \Big(\frac{1}{n!}\int_1^\infty t^{-1} e^{-t} (\log t)^n dt\Big) x^n.$$

Solution. Split $\Gamma(x) = \int_0^1 t^{x-1}e^{-t}dt + \int_1^\infty t^{x-1}e^{-t}dt$.

Step 1. (Over $(0, 1]$.) $R_N(t) = e^{-t} - \sum_{n \le N} \frac{(-t)^n}{n!}$ has $|R_N| \le \sum_{n>N} \frac{1}{n!} =: \rho_N \to 0$ on $[0, 1]$; with $\int_0^1 t^{n+x-1}dt = \frac{1}{n+x}$,

$$\Big|\int_0^1 t^{x-1}e^{-t}dt - \sum_{n \le N} \frac{(-1)^n}{n!(n+x)}\Big| = \Big|\int_0^1 t^{x-1} R_N(t)\, dt\Big| \le \frac{\rho_N}{x} \to 0.$$

Step 2. (Over $[1, \infty)$.) $t^{x-1}e^{-t} = t^{-1}e^{-t}e^{x\log t} = \sum_n \phi_n(t)$, $\phi_n(t) = \frac{x^n}{n!}t^{-1}e^{-t}(\log t)^n \ge 0$, each $\int_1^\infty \phi_n \le \int_1^\infty t^{x-1}e^{-t}$ finite.

Step 3. (Upper bound.) $\sum_{n\le N}\phi_n \le t^{x-1}e^{-t}$, so $\sum_n \int_1^\infty \phi_n \le \int_1^\infty t^{x-1}e^{-t}dt$.

Step 4. (Lower bound.) On $[1,c]$, $\log t \le \log c$, so $\sum \phi_n$ converges uniformly (Corollary 6.3) and (Proposition 6.5)

$$\int_1^c t^{x-1}e^{-t}dt = \sum_n \int_1^c \phi_n \le \sum_n \int_1^\infty \phi_n.$$

Let $c\to\infty$: with Step 3, $\int_1^\infty t^{x-1}e^{-t}dt = \sum_n \int_1^\infty \phi_n$, the second series. Add Step 1. ■